

# 20251103EB\_MN (seahorse, RNA)

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2025年11月3日 星期一

split iPSCs into a full 6-well plate on the Friday before differentiation

differentiate 5 of 6-wells with 10mL T2 in cell-repellent plates

keep some iPSCs

2025年11月5日 星期三

T2 -> T3

2025年11月7日 星期五

T3 -> T4 (~14 mL for over-weekend)

2025年11月10日 星期一

T4 -> T5

2025年11月12日 星期三

T5 -> T6

2025年11月14日 星期五

T6 -> T6 (~14mL for over-weekend)

2025年11月17日 星期一

T6 -> T7

Dissociation

1. Coat 0.01% Poly-L-Ornithine (PLO) Solution with sodium borate (1:100) for 4-5 hrs
2. Remove the PLO and wash the plate with PBS once
3. Coat the plate with laminine for 1hr.
4. Collect EBs in 50mL falcon and spin them down with 200rcf, 1min. Remove the supernatent (can leave some liquid to avoid remove EBs).
5. Add trypsin (usually 3-4 mL/10-cm plate) and pipette it up and down with 5mL serological pipette.
6. Put the tube into 37C water bath. Pipette the cells with P1000 every 5 min round 8-10 times until 15 min. (3 times in total. There might be more resistance at the 2nd round)
7. Add 2x volumn of Heat Inactivated FBS into the tube and pipette it with 5mL serological pipette gently. (Try to make the viscous cluster get through the tip to break it down)
8. Add 2x volumn of CTWM and pipette it with 5mL serological pipette gently.

9. Filter the cell with 40um filter into a new 50mL tube
10. Spin the cell down at 300rcf, 7min. Remove the supernatent.
11. Add T7 (~5mL) to resuspend the cells and count it.
12. Plate the cells with T7. (350-500K/well for 12well plate; 1-1.5M/well for 6-well plate)

cell info ^

|   | A    | B       |
|---|------|---------|
| 1 | cell | passage |
| 2 | WT11 |         |

Table1 ^

|   | A           | B                         |
|---|-------------|---------------------------|
| 1 | run1        | volume (mL)               |
| 2 | trypsin     | 3                         |
| 3 | HI-FBS/CTWM | 6                         |
| 4 | run2        | *take some undissolved EB |
| 5 | trypsin     | 2                         |
| 6 | HI-FBS/CTWM | 4                         |

Table3 ✓

|   | A            | B        | C            | D               | E      | F          | G                   | H               | I        |
|---|--------------|----------|--------------|-----------------|--------|------------|---------------------|-----------------|----------|
| 1 | conc. (M/mL) | survival | total volume | tota cell count | plate# | well plate | cell count/well (K) | number of wells | volume   |
| 2 | 4.26         | 99       | 7            | 29.82           | A      | 96         | 60                  | 36              | 507      |
| 3 |              |          |              |                 | B      | 24         | 250                 | 16              | 939      |
| 4 |              |          |              |                 | C,D    | 6          | 1972                | 12              | the rest |

| Plate A |   |   |   |   |   |   |   |   |   |    |    |    | ^ |
|---------|---|---|---|---|---|---|---|---|---|----|----|----|---|
|         | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 |   |
| A       |   |   |   |   |   |   |   |   |   |    |    |    |   |
| B       |   |   |   |   |   |   |   |   |   |    |    |    |   |
| C       |   |   |   |   |   |   |   |   |   |    |    |    |   |
| D       |   |   |   |   |   |   |   |   |   |    |    |    |   |
| E       |   |   |   |   |   |   |   |   |   |    |    |    |   |
| F       |   |   |   |   |   |   |   |   |   |    |    |    |   |
| G       |   |   |   |   |   |   |   |   |   |    |    |    |   |
| H       |   |   |   |   |   |   |   |   |   |    |    |    |   |

| Plate B |   |   |   |   | ^ |
|---------|---|---|---|---|---|
|         | 1 | 2 | 3 | 4 |   |
| A       |   |   |   |   |   |
| B       |   |   |   |   |   |
| C       |   |   |   |   |   |

2025年11月18日 星期二

Change T7 -> T7 if added virus

2025年11月19日 星期三

T7-> T7 if not added virus

2025年11月20日 星期四

T7 -> T8

2025年11月21日 星期五

1/2 T8 -> 1/2 T8

2025年11月24日 星期一

1/2 T8 -> 1/2 T8

2025年11月25日 星期二

Plate A

24h treatment: 13:20

Plate B

24h treatment: 16:30

2025年11月26日 星期三

1/2 T8 -> 1/2 T8

Plate A Seahorse

condition: 1.5uM oligomycin, 1uM FCCP, 0.5uM Rot/AA

Prepare drugs: 11:20-11:45

Change media at ~13:35

Incubate cells at 13:52

calibration at ~14:25-30

Put cells on machine at: 14:55

Assay finished at ~16:35

1. Rehydrate the cartridge the day before.
2. Turn on the seahorse machine.
3. Prepare drugs from stock as the drug table below with two glucose concentrations (with and without 1mM pyruvate)
4. every well wash with 70 uL media with corresponding glucose concentration.
5. add 175uL of media to the cells and add 2.5mM glucose media to all the blank wells.
6. incubate the plate for 1 hour to deplete CO2.
7. Take out the cartridge and remove the pink part.
8. Aliquot 148uL drugs into three 8-tube strips like below

| how I aliquot drug into 8-tube strips |         |            |             |            |             |                     |                      |                     |                      |
|---------------------------------------|---------|------------|-------------|------------|-------------|---------------------|----------------------|---------------------|----------------------|
|                                       | A       | B          | C           | D          | E           | F                   | G                    | H                   | I                    |
| 1                                     | oligo   | 2.5mM gluc | 0.15mM gluc | 2.5mM gluc | 0.15mM gluc | 2.5mM gluc+pytuvate | 0.15mM gluc+pytuvate | 2.5mM gluc+pytuvate | 0.15mM gluc+pytuvate |
| 2                                     | FCCP    | 2.5mM gluc | 0.15mM gluc | 2.5mM gluc | 0.15mM gluc | 2.5mM gluc+pytuvate | 0.15mM gluc+pytuvate | 2.5mM gluc+pytuvate | 0.15mM gluc+pytuvate |
| 3                                     | Rot/A A | 2.5mM gluc | 0.15mM gluc | 2.5mM gluc | 0.15mM gluc | 2.5mM gluc+pytuvate | 0.15mM gluc+pytuvate | 2.5mM gluc+pytuvate | 0.15mM gluc+pytuvate |

9. Use multiwell pipette with 4 tips and add the drugs to the cartridge as below. Add H2O for all the remaining wells.

| drug in cartridge layout_mito stress |            |          | ^ |
|--------------------------------------|------------|----------|---|
|                                      | A          | B        |   |
| 1                                    | A (oligo)  | B (FCCP) |   |
| 2                                    | C (Rot/AA) | D (H2O)  |   |

| drug calculation per glucose concentration |             |        |            |                     |                           |              |                           |                          |              |                |                           |                  |                   |                               |                          |                      |                    |                 |               |
|--------------------------------------------|-------------|--------|------------|---------------------|---------------------------|--------------|---------------------------|--------------------------|--------------|----------------|---------------------------|------------------|-------------------|-------------------------------|--------------------------|----------------------|--------------------|-----------------|---------------|
|                                            | A           | B      | C          | D                   | E                         | F            | G                         | H                        | I            | J              | K                         | L                | M                 | N                             | O                        | P                    | Q                  | R               | S             |
| 1                                          |             |        | stock (uM) | 1st dilution factor | total volume for 1st dil. | stock volume | media volume for 1st dil. | conc. after 1st dilution | working (uM) | loading volume | well volume before loaded | loading dilution | loading conc.(uM) | dilution factor from 1st dil. | well# for each condition | total loading volume | volume of 1st dil. | volume of media |               |
| 2                                          | mito stress | oligo  | 10000      | 10                  | 20                        | 2            | 18                        | 1000                     | 1.5          | 25             | 175                       | 8                | 12                | 83.3333333333                 | 10                       | 300                  | 3.6                | 296.4           |               |
| 3                                          |             | FCCP   | 10000      | 20                  | 20                        | 1            | 19                        | 500                      | 1            | 25             | 200                       | 9                | 9                 | 55.5555555556                 | 10                       | 300                  | 5.4                | 294.6           |               |
| 4                                          |             | Rot/AA | 10000      | 10                  | 10                        | 1            | 8                         | 1000                     | 0.5          | 25             | 225                       | 10               | 5                 | 200                           | 10                       | 300                  | 1.5                | 297.0           | one condition |
| 5                                          |             | medium |            |                     |                           |              |                           |                          |              | 275            |                           |                  |                   | total occupied wells (Mito)   | 10                       | 3300                 |                    | 6600            | 4188          |
| 6                                          |             |        |            |                     |                           |              | 45                        |                          |              |                |                           |                  |                   |                               | 10440                    |                      |                    | 7488            |               |
| 7                                          |             |        |            |                     |                           |              |                           |                          |              |                |                           |                  |                   |                               | 17973                    |                      |                    | 7533            |               |

use 2mM glutamine for this time (instead of 4mM)

| prepare seahorse basal media |           |            |              |                  |                      |                 |         |
|------------------------------|-----------|------------|--------------|------------------|----------------------|-----------------|---------|
|                              | A         | B          | C            | D                | E                    | F               | G       |
| 1                            |           | stock (mM) | working (mM) | loading dilution | total loading volume | volume of media |         |
| 2                            | glucose   | 1100       | 2.5          | 440              | 20000                | 45.5            | 19754.5 |
| 3                            |           | 1100       | 0.15         | 7333.33333333    | 8500                 | 1.2             | 8413.8  |
| 4                            | glutamine | 200        | 2            | 100              | 20000                | 200.0           |         |
| 5                            |           | 200        | 2            | 100              | 8500                 | 85.0            |         |
| 6                            | pyruvate  | 100        | 1            | 100              | 4200                 | 42.0            | 4158    |

| Cell conditions |   |                            |   |   |                            |   |                            |   |   |    |    |    |
|-----------------|---|----------------------------|---|---|----------------------------|---|----------------------------|---|---|----|----|----|
|                 | 1 | 2                          | 3 | 4 | 5                          | 6 | 7                          | 8 | 9 | 10 | 11 | 12 |
| A               |   |                            |   |   |                            |   |                            |   |   |    |    |    |
| B               |   |                            |   |   |                            |   |                            |   |   |    |    |    |
| C               |   |                            |   |   |                            |   | cells attached a bit weird |   |   |    |    |    |
| D               |   |                            |   |   | cells attached a bit weird |   |                            |   |   |    |    |    |
| E               |   |                            |   |   |                            |   |                            |   |   |    |    |    |
| F               |   | cells attached a bit weird |   |   |                            |   |                            |   |   |    |    |    |
| G               |   |                            |   |   |                            |   |                            |   |   |    |    |    |
| H               |   |                            |   |   |                            |   |                            |   |   |    |    |    |

| assay map |   |             |   |   |   |   |                                 |   |   |    |    |    | ^ |
|-----------|---|-------------|---|---|---|---|---------------------------------|---|---|----|----|----|---|
|           | 1 | 2           | 3 | 4 | 5 | 6 | 7                               | 8 | 9 | 10 | 11 | 12 |   |
| A         |   |             |   |   |   |   |                                 |   |   |    |    |    |   |
| B         |   |             |   |   |   |   |                                 |   |   |    |    |    |   |
| C         |   | 2.5mM mito  |   |   |   |   | 2.5mM mito (with 1mM pyruvate)  |   |   |    |    |    |   |
| D         |   | 0.15mM mito |   |   |   |   | 0.15mM mito (with 1mM pyruvate) |   |   |    |    |    |   |
| E         |   | 2.5mM mito  |   |   |   |   | 2.5mM mito (with 1mM pyruvate)  |   |   |    |    |    |   |
| F         |   | 0.15mM mito |   |   |   |   | 0.15mM mito (with 1mM pyruvate) |   |   |    |    |    |   |
| G         |   |             |   |   |   |   |                                 |   |   |    |    |    |   |
| H         |   |             |   |   |   |   |                                 |   |   |    |    |    |   |

extract genomic DNA using PureLink™ Pro 96 Genomic DNA Purification Kit for normalization  
see [notes](#)

Plate B (RNA extraction)

| Well1 |                |   |                 |   | ^ |
|-------|----------------|---|-----------------|---|---|
|       | 1              | 2 | 3               | 4 |   |
| A     | 2.5mM gluc 1h  |   | 2.5mM gluc 24h  |   |   |
| B     | 0.15mM gluc 1h |   | 0.15mM gluc 24h |   |   |
| C     |                |   |                 |   |   |

Treat cells at 19:05  
Collect cells with 200uL Trizol reagent/well at 20:05 (3.5h more for 24h)  
freeze the sample in -80C

2025年12月24日 星期三

Plate B (RNA extraction) cont' (RNA extraction, RT and qPCR)  
<https://benchling.com/s/etr-GBT2cZpFYDohY81g7XgV?m=slm-HyXmqSVuvTooNuimP5XN>